



# Constructing a High-Resolution 3D MRI-Based Brain Atlas of Octodon degus for Aging and Alzheimer's Disease Research

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## Abstract

Alzheimer's disease (AD) is the most common form of dementia, yet effective treatments remain limited and no definitive cure exists. Although mice are the most widely used animal model in neuroscience research, they do not naturally develop the spectrum of pathologies characteristic of human AD, limiting their translational relevance. The Octodon degus, a long-lived, diurnal rodent that can spontaneously develop human AD-like neuropathologies, offers a promising natural model for studies of aging, neurodegeneration, and Alzheimer's disease. However, the utilization of degus in neuroscience research has been constrained by the shortcomings of existing degu brain atlases. These preliminary degu atlases lack reliable stereotaxic coordinates, suffer from low spatial resolution due to large inter-slice intervals, provide limited multi-planar (coronal, sagittal, and horizontal) representations, and offer incomplete anatomical delineation and labeling of brain regions and subregions. Together, these limitations hinder experimental precision and reproducibility in degu brain studies.

Here we present our progress in the construction of a high-resolution 3D atlas of the degu brain. High-field magnetic resonance imaging (MRI) was performed at the University of Calgary using a 9.4T MRI system achieving an isotropic voxel resolution of 50  $\mu$ m. To achieve detailed anatomical parcellation, the degu MRI datasets were registered onto the established rat Waxholm atlas, generating a high-resolution, annotated atlas adapted for the degu brain. An additional group of young and aged animals underwent in-vivo micro-computed tomography ( $\mu$ CT) imaging to characterize age-related skull morphology and complement our atlas.

Future work will further enhance our atlas by integrating light-sheet or serial two-photon tomography of the degu brain with tissue clearing and immunolabeling, enabling incorporation of cell-type-specific molecular information. This multimodal framework will produce the most comprehensive and anatomically precise degu brain atlas to date, facilitating wider adoption of Octodon degus as a critical model for neuroscience, aging, and Alzheimer's disease research.

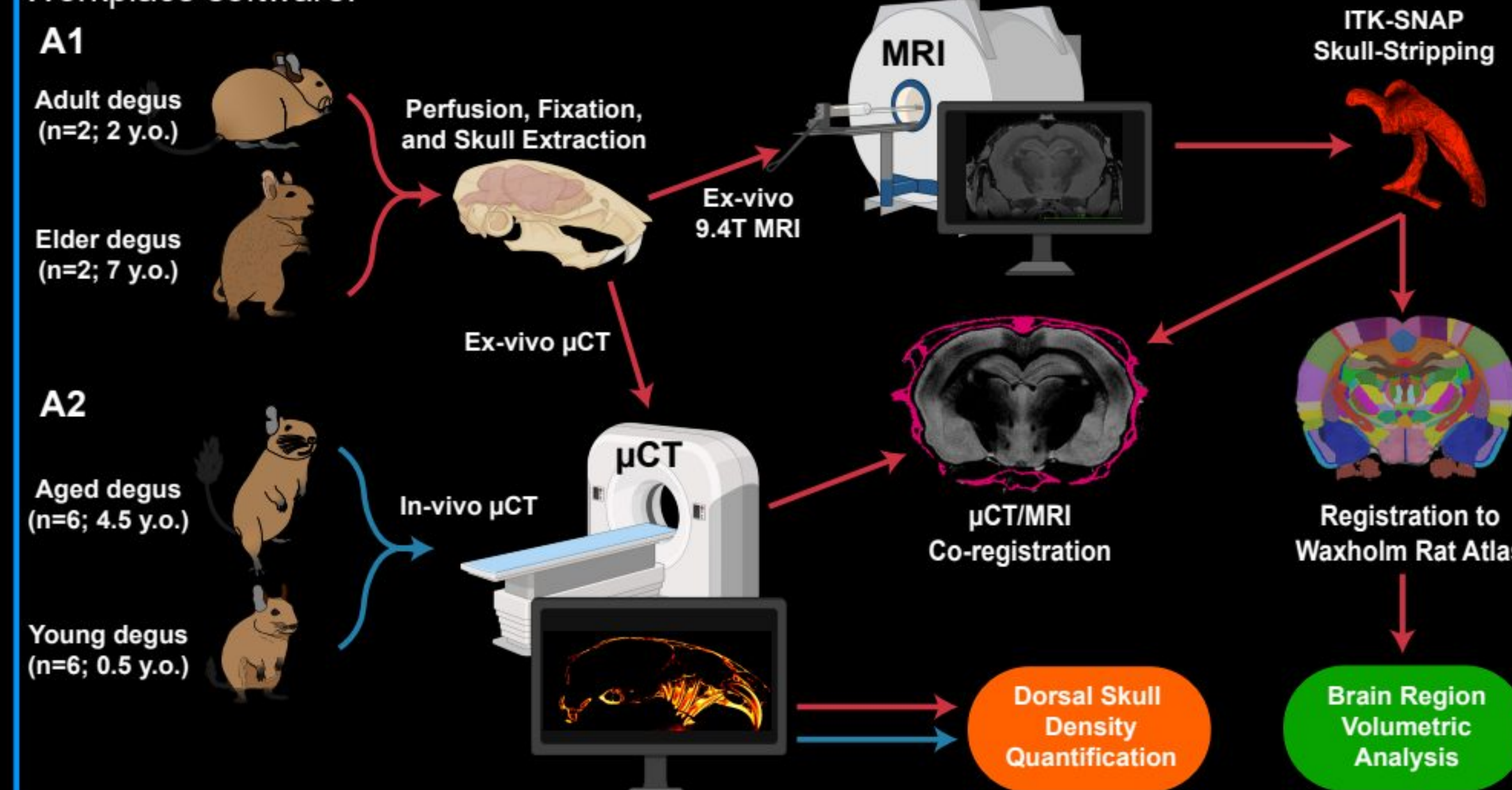
## Background & Methods

**Background:** Brain atlases are crucial tools for neuroscience researchers studying brain anatomy, neurological development/plasticity, and neurodegenerative disorders. This vital tool allows researchers to accurately target and perform experiments on precise areas of the brain.

This project studies the *Octodon degus* (degu) a long-lived rodent endemic to Chile. Degus possess many qualities that make them excellent models for translational neuropathological research, as they can naturally develop pathologies such as amyloid- $\beta$  plaques, neurofibrillary tangles, neuroinflammation, and neuron loss, all of which are present in human Alzheimer's disease. Other factors that make them a valuable models are their diurnality, social behaviors, extended lifespan, and their propensity to develop diabetes and atherosclerosis.

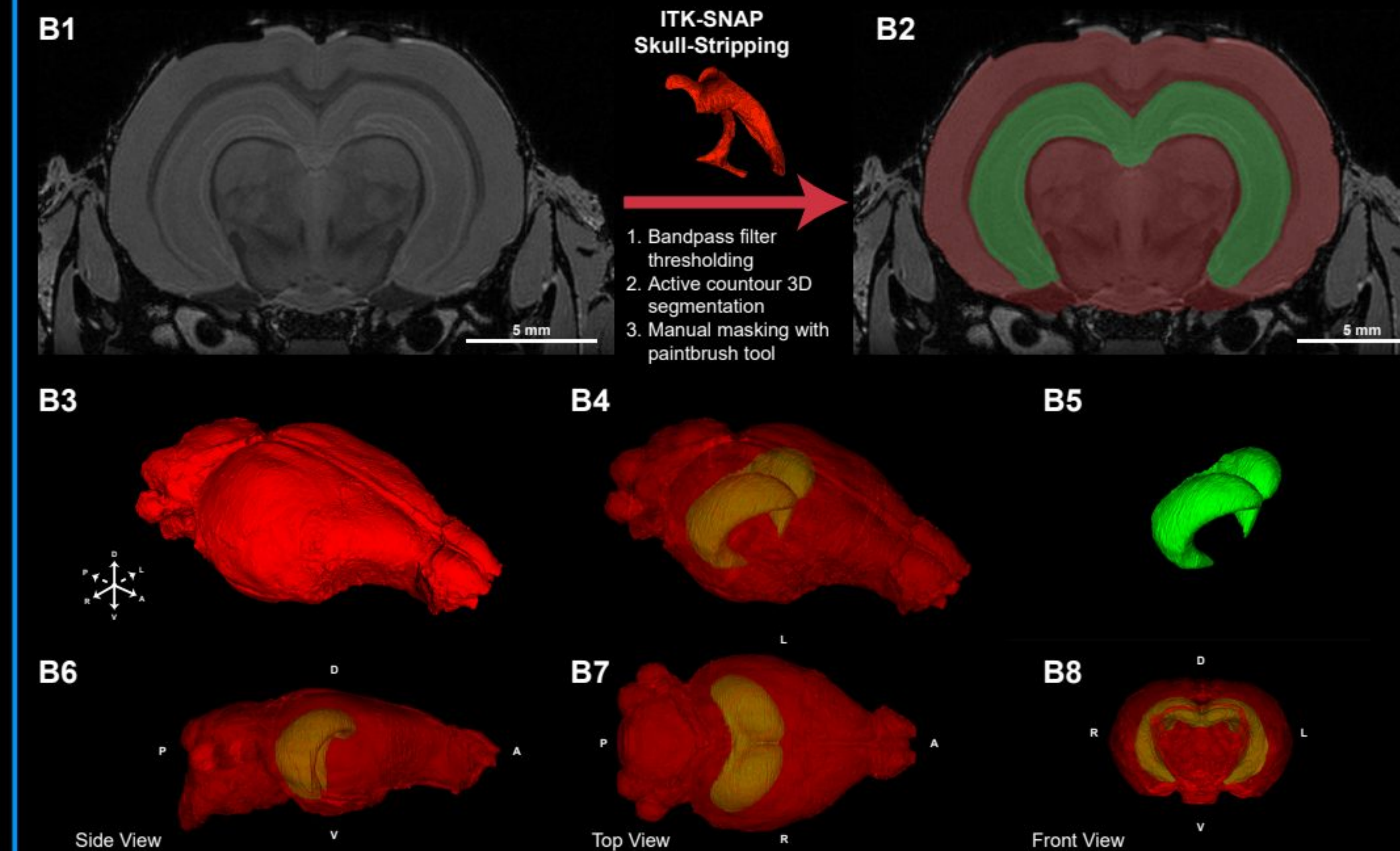
**Methods:** **A1.** Brains were obtained from wild-caught, outbred Chilean degus ( $n=4$ ), spanning adult and advanced age cohorts. Animals were transcardially perfused, and intact heads were post-fixed in paraformaldehyde prior to imaging to preserve anatomical integrity. High-field magnetic resonance imaging (MRI) was performed using a 9.4T MRI system achieving an isotropic voxel resolution of 50  $\mu$ m. Following image acquisition, skull stripping was performed using ITK-SNAP software. Degu MRI datasets were registered onto the established rat Waxholm atlas to achieve detailed anatomical parcellation, generating a high-resolution, annotated atlas adapted for the degu brain. Brain region volumetric analysis was performed using ITK-SNAP.

**A2.** The outbred degu skulls from (A) and a cohort of live young and aged degus ( $n=12$ ) underwent  $\mu$ CT imaging. Scans were visualized and analyzed using the Inveon Research Workplace software.

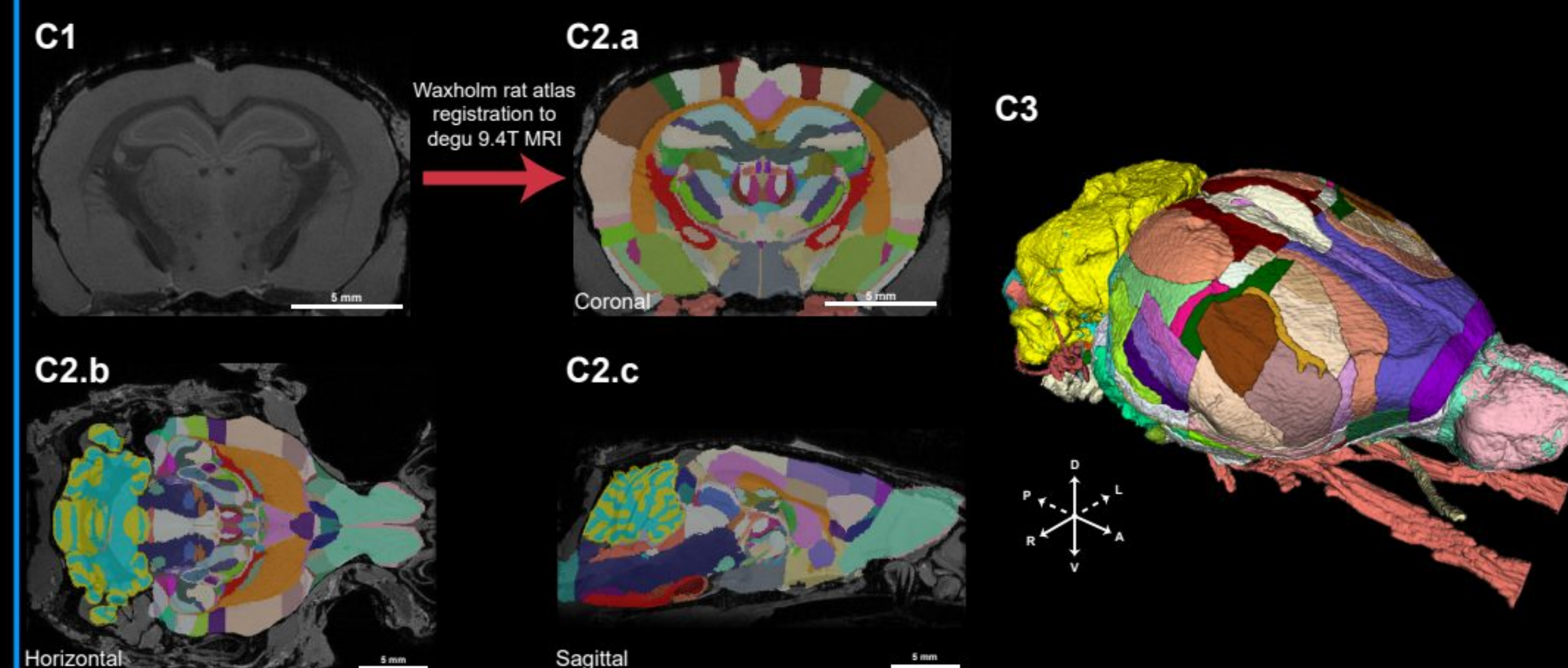


## Results - MRI

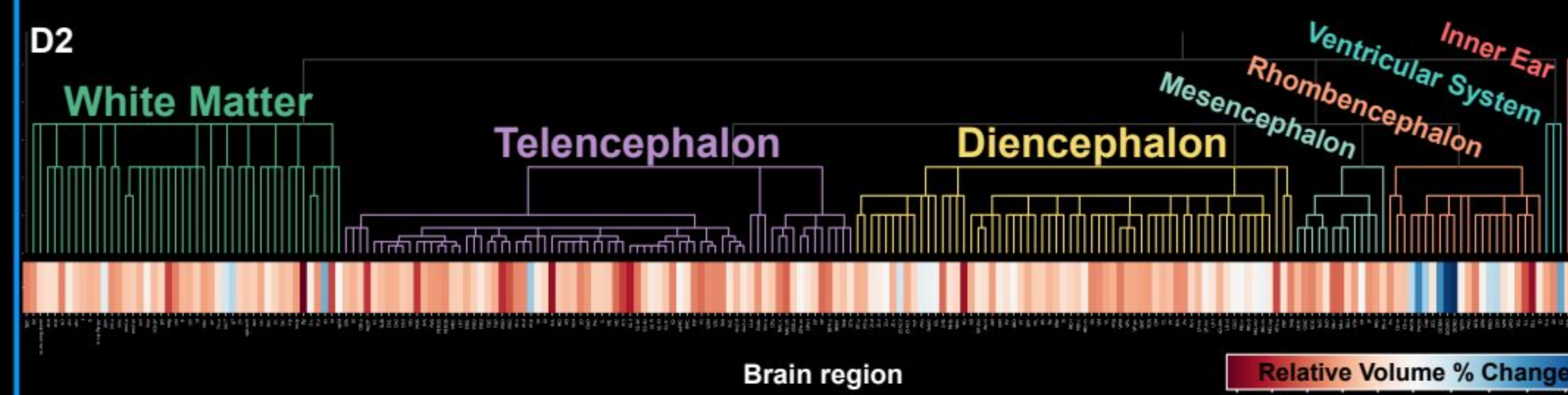
B. Skull stripping (removal of skull, muscles, and other tissues) of 9.4T MRI scans using the active contour 3D segmentation tool and bandpass thresholding filter from ITK-SNAP 3D isofware (brain in red, hippocampus in green). The resulting 3D masks were fed into the atlas registrations pipeline.



C. Registration of Waxholm rat atlas to degu MRI scans. Three anatomical planes and 3D render are shown in C2-C3 showing the 222 labeled brain regions.

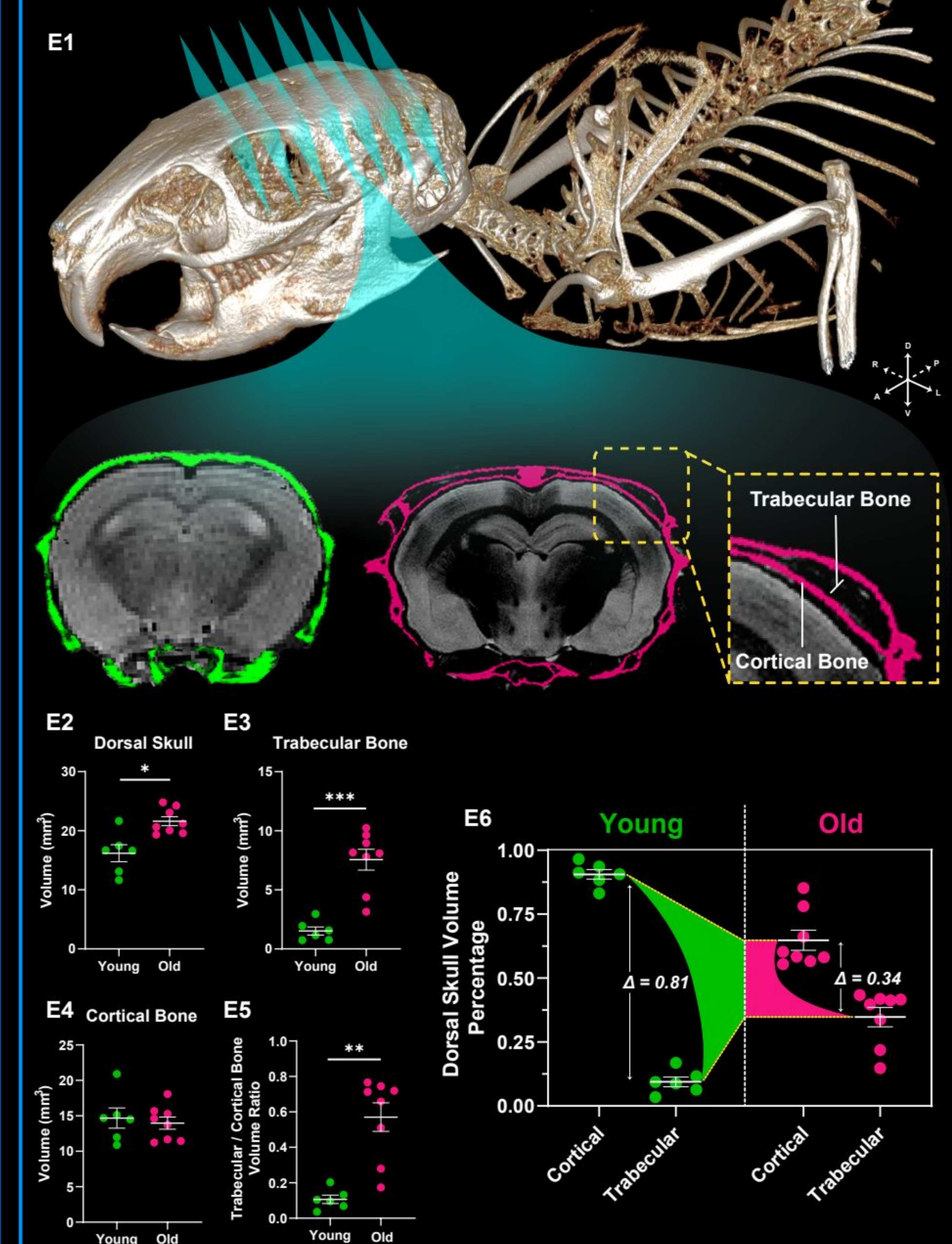


D. Preliminary volumetric analysis of registered degu MRI scans suggests old degus exhibit broad brain volume reduction. D1. From 222 brain regions, 152 show greater than 5% volume loss, while only 13 show greater than 5% increase. Regions with greatest volume loss seem to be larger brain regions. D2. Hierarchical brain regions dendrogram (based on the Waxholm rat atlas) shows most brain regions exhibiting volume loss occur in the rhombencephalon of the degu brain.



## Results - $\mu$ CT

E. In-vivo  $\mu$ CT analysis shows aged degus develop an expansion of trabecular skull bone. E1. Representative  $\mu$ CT scans of the degu upper body skeleton and coronal views of their skull co-registered with MRI brain scans. Young degus in green, old in magenta, with a zoom-in showing the trabecular bone expansion. E2-E6 Volumetric analysis of the degu dorsal skull (E2), trabecular bone (E3), cortical bone (E4), and their ratios/volumetric percentages (E5-E6) show aged degus possess increased trabecular bone composition in their skulls.



## Conclusions

1. Using the Waxholm rat atlas framework, we have successfully mapped 222 regions of the degu brain in young and old animals, laying the groundwork for our novel brain atlas.
2. Preliminary volumetric quantification suggests aged degus exhibit shrinkage of numerous brain regions when compared to young degus.
3. Aged degus develop an expansion of trabecular/spongy bone in their dorsal skull.

## Acknowledgements & References

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